MIL-B-13393E(EA)
13 June 1986
SUPERSEDING
MIL-B-13393D
6 January 1972

MILITARY SPECIFICATION

BOMB, SMOKE OR INCENDIARY, AN-M47A4, 100-LB; INERT COMPONENTS FOR

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the components for one type of 100-pound incendiary or smoke bomb.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

VV-L-800 - Lubricating Oil, General Purpose, Preservative, Water Displacing, Low Temperature.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant (For Carton Sealing).
PPP-B-636 - Box, Fiberboard.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1325

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-C-23411 - Corrosion Preventative Compound, Clear.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Number of Ammunition.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 14-5-645 - Bomb, 100-Pound, AN-M47A4; Metal Parts.
5220-A50-9059A - Plug, Taper Thread.
5220-A50-9062 - Ring, Taper Thread and Plug, Feeler, Double End.
5220-A50-9061 - Plug, Taper Check.

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 14-5-645 and on the subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.2 Assembly. The bomb components shall be assembled as shown on Drawing 14-5-645.

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3.3 Leakage and distortion.

3.3.1 Bomb body assembly. The bomb assembly shall not leak nor be distorted when subjected to an internal hydrostatic pressure of 300 - 25 pounds per square inch gage (psig) for a period of 30 + 5 seconds after the specified pressure has been reached, when tested as specified in 4.4.4.1(a).

3.3.2 Burster well. The burster well shall not leak nor be distorted when subjected to an internal hydrostatic pressure of 300 - 25 psig for a period of 30 + 5 seconds after the specified pressure has been reached, when tested as specified in 4.4.4.1(b).

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of components shall be produced in accordance with this specification for examination and test.

3.5 Workmanship. The components shall be free of all foreign matter (oil, grease, contamination, etc.) and damage such as dents, cracks, or deformation which could affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of components sufficient to assemble 10 bombs and 10 burster well assemblies shall be produced, using the same methods, materials, and equipment that will be used in regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. All components, subassemblies and assemblies of the preproduction sample shall be examined for dimensional accuracy in accordance with the applicable drawings.

4.3.2.2 For test. Ten each, bomb body assemblies and burster well assemblies shall be tested in accordance with 4.4.4.1(a) and 4.4.4.1(b) respectively.

4.3.3 Acceptance/rejection criteria. Preproduction samples shall meet all the requirements of this specification to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall be identified and controlled in accordance with MIL-STD-1168 and consist of the same type components, of one interfix number, produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for delivery at one time.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. Sample components shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For critical defects. Each bomb body assembly and burster well in the lot shall be inspected for the critical characteristics listed in the classification of defects.

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4.4.3.3 Classification of defects.

(a) Bomb, 100-lb, AN-M47A4 (Dwg 14-5-645).

Categories	Defects	Inspection equipment	Acceptance standards
Critical:			
1	Leakage or distortion		4.4.4.1(a)
Major:	AQL 1.0 percent defective		
101	Component missing		
102	Component incorrectly assembled		
103	Outside diameter incorrect (after leak test)		
104	Misalignment of fin to body		
105	Length incorrect		
106	Lug location incorrect		
107	Lug hole incorrect (Dwg 14-5-304)		
108	Lug radius incorrect (Dwg 14-5-304)		
109	Distance between lugs incorrect		
110	Suspension band height (10-1/4) incorrect		
111	Thread incorrect (Dwg 14-5-308)	Plug, Taper Thread (Dwg 5220-A50-9059A)	
112	Workmanship		
Minor:	AQL 4.0 percent defective		
201	Protective finish missing		
202	Protective finish incorrect		
203	Marking missing or incorrect		
204	Marking illegible		

(b) Spacer (Dwg 14-5-400).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter incorrect
102	Inside diameter incorrect
103	Thickness incorrect
104	Angle incorrect
105	Workmanship (3.5)

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(c) Well, burster assembly (Dwg 14-5-225).

Categories	Defects	Inspection equipment	Acceptance standards
Critical:			
1	Leakage or distortion		4.4.4.1(b)
Major:	AQL 1.0 percent defective		
101	Chamber excessive		
102	Thread (1-1/2-12UNF-2B) incorrect		
103	Fuze adapter inside diameter (1.240) incorrect		
104	Overall length (max) incorrect		
105	Thread cavity depth (including undercut) incorrect		
106	Depth (0.91 max) incorrect		
107	Body outside diameter incorrect (after leakage)		
108	Burster Cavity inside diameter incorrect		
109	Burster cavity depth incorrect		
110	External pipe thread incorrect	Ring (Dwg 5220-A50-9062) Plug (Dwg 5220-A50-9061)	
111	Plating missing		
112	Workmanship (3.5)		

(d) Fin assembly (Dwg 14-5-649 or 14-5-888).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing
102	Component incorrectly assembled
103	Support outside diameter (1-1/2) incorrect
104	Support location incorrect
105	Fin side profile incorrect
106	Distance diagonally across fins (max) incorrect
107	Distance between adjacent fins (max) incorrect
108	Spot welds
109	Workmanship (3.5)

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Minor: AQL 2.5 percent defective

*201 Support hole diameter or location incorrect

*To be checked prior to assembly

(e) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Interior of nose, body and fin cone not preserved
102	Loose pack (insufficient filler pads)
103	Boxes not properly closed
104	Strapping loose or missing
105	Marking incorrect
Minor:	AQL 10.0 percent defective
201	Marking illegible

4.4.4 Tests.

4.4.4.1 Leakage and distortion.

(a) Bomb body assembly. The empty body assembly, less the burster well, shall be subjected to the hydrostatic pressure for the time as specified. Exudation or failure to hold the specified pressure indicates presence of a leak and shall be cause for rejection. Permanent distortion or rupture of bomb shall be cause of rejection.

(b) Burster well. The burster well shall be subjected to the hydrostatic pressure for the time as specified. Exudation or failure to hold the specified pressure indicates presence of a leak and shall be cause for rejection. Permanent distortion or rupture of burster well shall be cause of rejection.

5. PACKAGING

5.1 Cleaning, drying and preservation.

5.1.1 Cleaning and drying. The interior of the bomb assembly, Drawing D14-5-645, and the burster well shall be cleaned and dried in accordance with MIL-P-116. Cleaning process C-1 and any applicable drying process shall be used.

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5.1.2 Preservation. The bomb assembly, cleaned and dried in accordance with 5.1.1 shall have the interior surfaces of the nose, body and fin cone coated with preservative oil conforming to VV-L-800. Excess oil shall be drained off. The burster well shall have interior threaded surfaces coated with a light coat of preservative compound conforming to type I or II, MIL-C-23411. The open end shall have a plastic can-plug installed to close the opening. Ship plastic plugs separately if burster wells are being shipped directly to the loading facility. The external threads shall be covered with a wrap of barrier material conforming to tpe I or II, MIL-B-121, which shall be secured in place by tape conforming to PPP-T-76.

5.1.3 Packaging, level C. The burster well shall be packaged, nine per box in fiberboard boxes conforming to style FOL, type CF, class domestic, variety SW, grade 200 min, PPP-B-636, having inside dimensions approximating 6 by 6 by 40 inches in length, width and depth respectively. The burster wells shall be separated from each other by means of separator pads fabricated of domestic class fiberboard, including additional pads or fillers, as necessary, to restrict free movement of the items. The boxes shall be closed in accordance with method II of the appendix to PPP-B-636.

5.2 Packing, level C.

5.2.1 Bomb assembly. The bomb assemblies, Dwg D14-5-645, preserved in accordance with 5.1.2, shall be packed in accordance with Dwg D14-17-270 except as otherwise specified herein. The arming wire and data card are not included. The box cover is secured by strapping only, excluding any nailing in closing the box. The strapping may be class A or C for flat strapping or finish $\frac{1}{\Lambda}$ for round strapping.

5.2.2 Burster well. The burster wells, packaged in accordance with 5.1.3, shall be packed 36 (4 packages) per box in wirebound boxes conforming to class 1, style optional, 125 to 200 lb load limit, PPP-B-585, having inside dimensions of 41 by 13 by 13 inches length, width and depth, respectively. The boxes shall be closed and strapped in accordance with the appendix to PPP-B-585, including suitable filler pads, as necessary to restrict free movement of the packages.

5.3 Marking. In conjunction with information furnished in the contract or order, boxes shall be marked in accordance with applicable requirements of MIL-STD-129 utilizing labels or tags. Address marking is not required on individual boxes, when palletized.

5.4 Palletization. When specified in the contract or order, bomb assemblies, packed in accordance with 5.2, shall be palletized in accordance with Drawing D14-5-2827, except that two unitizing straps and marking label are not required. Each palletized load shall be provided with a label or tag showing quantity, item name and lot number and gross weight and cube of load. Address marking is not required on palletized unit loads when shipped in full carload of full truckload lots moving from a single consignor to a single consignee.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Any special marking required.
(c) Preproduction.

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1325-A477

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-13393E(EA)	BOMB, SMOKE OR INCENDIARY, AN-M47A4, 100-LB; INERT COMPONENTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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